Name :- Mihir Kumar
Regd no. :- 2141011109

User Space

This is where user-level applications, & process execute. These applications interact with the kernel through system calls to request resources & services

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System Call Interface

This interface acts as a crucial boundary between user space and kernel space. It provides a standardized way for user applications to request kernel services

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Memory Management

This subsystem handles the allocation and deallocation of the system's memory resources, it manages virtual memory & handles swapping between RAM and disk

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Virtual File System (VFS)

This VFS provides a consistent interface for accessing different types of files systems, it abstracts the underlying file system implementation details.

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Network Stack

This component implements various networking protocols (TCP/IP) that enable communication over a network. It handles packet routing and network connections.

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Inter-Process Communication (IPC)

IPC mechanisms allow different process to communicate and synchronize with each other, common IPC techniques include poles, queue & shared memory

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Security Modules (eg, SELinux, AppArmor)

These are kernel frameworks that implement security policies to control access to system resources. They enhance system security through MAC.

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Device Drivers

This represents the physical components the computer system, such as CPU, memory, storage devices & peripherals. The kernel directly interacts with hardware.